

Exercise Solution

problem 3.1: Propositional Logic for the Hazardous Warehouse

From [Problems](#)

Grid layout (for reference):

```
(1,3) -- (2,3) -- (3,3)
  |       |       |
(1,2) -- (2,2) -- (3,2)
  |       |       |
(1,1) -- (2,1) -- (3,1)
```

1. Physics encoding

Creaking rules — 9 biconditionals, one per square. Creaking at (i, j) iff damaged floor in at least one adjacent square:

$$C_{1,1} \Leftrightarrow D_{2,1} \vee D_{1,2}$$

$$C_{2,1} \Leftrightarrow D_{1,1} \vee D_{3,1} \vee D_{2,2}$$

$$C_{3,1} \Leftrightarrow D_{2,1} \vee D_{3,2}$$

$$C_{1,2} \Leftrightarrow D_{1,1} \vee D_{2,2} \vee D_{1,3}$$

$$C_{2,2} \Leftrightarrow D_{2,1} \vee D_{1,2} \vee D_{3,2} \vee D_{2,3}$$

$$C_{3,2} \Leftrightarrow D_{3,1} \vee D_{2,2} \vee D_{3,3}$$

$$C_{1,3} \Leftrightarrow D_{1,2} \vee D_{2,3}$$

$$C_{2,3} \Leftrightarrow D_{2,2} \vee D_{1,3} \vee D_{3,3}$$

$$C_{3,3} \Leftrightarrow D_{2,3} \vee D_{3,2}$$

Rumbling rules — 9 biconditionals with the same structure, replacing C with N and D with F :

$$\begin{aligned} R_{1,1} &\Leftrightarrow F_{2,1} \vee F_{1,2} \\ R_{2,1} &\Leftrightarrow F_{1,1} \vee F_{3,1} \vee F_{2,2} \\ R_{3,1} &\Leftrightarrow F_{2,1} \vee F_{3,2} \\ R_{1,2} &\Leftrightarrow F_{1,1} \vee F_{2,2} \vee F_{1,3} \\ R_{2,2} &\Leftrightarrow F_{2,1} \vee F_{1,2} \vee F_{3,2} \vee F_{2,3} \\ R_{3,2} &\Leftrightarrow F_{3,1} \vee F_{2,2} \vee F_{3,3} \\ R_{1,3} &\Leftrightarrow F_{1,2} \vee F_{2,3} \\ R_{2,3} &\Leftrightarrow F_{2,2} \vee F_{1,3} \vee F_{3,3} \\ R_{3,3} &\Leftrightarrow F_{2,3} \vee F_{3,2} \end{aligned}$$

Safety rules — 9 biconditionals:

$$S_{i,j} \Leftrightarrow \neg D_{i,j} \wedge \neg F_{i,j} \quad \text{for each } (i,j) \in \{1, 2, 3\}^2$$

Total: 27 propositional sentences encode the physics.

2. Initial knowledge

Starting square is safe:

$$S_{1,1}$$

which, combined with the safety biconditional, entails $\neg D_{1,1} \wedge \neg F_{1,1}$.

At least one square has damaged floor:

$$D_{1,1} \vee D_{2,1} \vee D_{3,1} \vee D_{1,2} \vee D_{2,2} \vee D_{3,2} \vee D_{1,3} \vee D_{2,3} \vee D_{3,3}$$

At least one forklift — same structure with F :

$$F_{1,1} \vee F_{2,1} \vee F_{3,1} \vee F_{1,2} \vee F_{2,2} \vee F_{3,2} \vee F_{1,3} \vee F_{2,3} \vee F_{3,3}$$

Total initial knowledge: $1 + 1 + 1 = 3$ sentences.

3. Scenario reasoning

Percepts at $(1, 1)$:

$$\neg C_{1,1}, \quad \neg R_{1,1}$$

Step 1: From $\neg C_{1,1}$ and the creaking biconditional $C_{1,1} \Leftrightarrow D_{2,1} \vee D_{1,2}$:

$$\neg C_{1,1} \Rightarrow \neg(D_{2,1} \vee D_{1,2}) \Rightarrow \neg D_{2,1} \wedge \neg D_{1,2}$$

Step 2: From $\neg R_{1,1}$ and the rumbling biconditional $R_{1,1} \Leftrightarrow F_{2,1} \vee F_{1,2}$:

$$\neg R_{1,1} \Rightarrow \neg(F_{2,1} \vee F_{1,2}) \Rightarrow \neg F_{2,1} \wedge \neg F_{1,2}$$

Step 3: Apply the safety biconditionals to the newly cleared squares:

- $(2, 1): \neg D_{2,1} \wedge \neg F_{2,1} \Rightarrow S_{2,1} \checkmark$
- $(1, 2): \neg D_{1,2} \wedge \neg F_{1,2} \Rightarrow S_{1,2} \checkmark$

Step 4: For all other squares — $(3, 1), (2, 2), (3, 2), (1, 3), (2, 3), (3, 3)$ — we have no information about either damage or forklift presence. Their safety cannot be determined.

Provably safe squares: $\{(1, 1), (2, 1), (1, 2)\}$.

Grid after observation at $(1, 1)$:

?	?	?	$(1, 3)$	$(2, 3)$	$(3, 3)$
S	?	?	$(1, 2)$	$(2, 2)$	$(3, 2)$
S	S	?	$(1, 1)$	$(2, 1)$	$(3, 1)$

4. Exploration

New percepts at $(2, 1)$:

$$C_{2,1}, \quad \neg R_{2,1}$$

Step 1: From $C_{2,1}$ and $C_{2,1} \Leftrightarrow D_{1,1} \vee D_{3,1} \vee D_{2,2}$:

$$D_{1,1} \vee D_{3,1} \vee D_{2,2}$$

We already know $\neg D_{1,1}$ (starting square is safe), so:

$$D_{3,1} \vee D_{2,2}$$

Step 2: From $\neg R_{2,1}$ and $R_{2,1} \Leftrightarrow F_{1,1} \vee F_{3,1} \vee F_{2,2}$:

$$\neg F_{1,1} \wedge \neg F_{3,1} \wedge \neg F_{2,2}$$

Step 3: Forklift exclusions so far: $\neg F_{1,1}, \neg F_{2,1}, \neg F_{1,2}, \neg F_{3,1}, \neg F_{2,2}$. The at-least-one-forklift constraint then tells us the forklift is at one or more of $(3, 2), (1, 3), (2, 3), (3, 3)$.

Step 4: Check safety of each unknown square:

Square	Damage?	Forklift?	Safe?
(3, 1)	Possible ($D_{3,1} \vee D_{2,2}$)	$\neg F_{3,1}$	Unknown
(2, 2)	Possible ($D_{3,1} \vee D_{2,2}$)	$\neg F_{2,2}$	Unknown
(3, 2)	Unknown	Possible	Unknown
(1, 3)	Unknown	Possible	Unknown
(2, 3)	Unknown	Possible	Unknown
(3, 3)	Unknown	Possible	Unknown

No individual square is provably dangerous. However, we know $D_{3,1} \vee D_{2,2}$, so at least one of (3, 1) and (2, 2) has damaged floor. Note that without an "exactly one" constraint, we cannot rule out damage at other unvisited squares either.

Grid after observation at (2, 1):

?	?	?	(1, 3)	(2, 3)	(3, 3)
S	?	?	(1, 2)	(2, 2)	(3, 2)
S	S	?	(1, 1)	(2, 1)	(3, 1)

Key new knowledge: at least one of (3, 1) and (2, 2) has damaged floor; the forklift is somewhere among the uncleared squares. The next useful move would be to explore (1, 2) — its percepts would help narrow down the remaining locations.

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